EMCH 290 HW#3

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1) a. H_2O $p = 100\ lb_f/in^2$

$u = 500\ Btu/lb$

$T = \boxed{327.86\ (°F)}$ from table A-3E

$v = \boxed{1.121\ (ft^3/lb)}$

$h = \boxed{520\ (Btu/lb)}$

$u = u_f + x(u_g - u_f)$

$(500) = (298.3) + x(1105.8 - 298.3)$

$\rightarrow x = 0.2498$

$v = v_f + x(v_g - v_f)$

$= (0.01774) + (0.2498)(4.434 - 0.01774)$

$= \boxed{1.121\ ft^3/lb}$

$h = h_f + x(h_g - h_f)$

$= (298.6) + (0.2498)(1187.8 - 298.6)$

$= \boxed{520.7}$

2) H_2O in closed, rigid tank, State 1 = sat. vapor

$\hookrightarrow \Delta m = 0$ $\hookrightarrow \Delta V = 0$

$T_1 = 180°C$

$T_2 = 100°C$

Find P_1 (MPa) & X_2

a. State 1:

- sat. vapor
- $T_1 = 180°C$
- $P_1 = 10$ bar $= 10E5\ Pa = \boxed{1\ MPa}$ from table

State 2:

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- $T_2 = 100°C$
- $P_2 = ?$

b. $\frac{a_1 - a_2}{a_1 - a_x} = \frac{b_1 - b_2}{b_1 - b_x}$

$\frac{T_1 - T_2}{T_1 - T_x} = \frac{P_1 - P_2}{P_1 - P_x}$

$\frac{99.63 - 111.4}{99.63 - (100)} = \frac{1.00 - 1.50}{1.00 - P_x}$

$P_x =$

x	y
99.63	1.0
111.4	1.5

$y = y_1 + (x - x_1)\frac{(y_2 - y_1)}{(x_2 - x_1)}$

$y = (1.0) + (100 - 99.63)\frac{(1.5 - 1.0)}{(111.4 - 99.63)}$

$P_2 = 1.0157$ bar

$PV = mRT$

$P\frac{V}{m} = RT$

$Pv = RT$

$v_2 = \frac{RT_2}{P_2}$

$= \frac{(8.314\ J/molK)(100°C + 273.15)K}{(1.0157E5)\ Pa}$

$= 0.0305\ m^3/kg$

$v = v_f + x(v_g - v_f)$

$(0.0305) = (v_f) + x(v_g - v_f)$

*interpolate to get v_f & v_g gives:

$v_f = 1.0435E-3$

$v_g = 1.6779$

$(0.0305) = (1.0435E-3) + x(1.6779 - 1.0435E-3)$

$\boxed{x = 0.01757}$

6/8

3) 2kg H_2O vapor expands constant pressure = 300 kPa (3 bar)

State 1

$P_1 = 3\,MPa$

$v_1 = v_g$

State 2

saturated vapor

$P_2 = 3\,MPa$

$V_2 = 2.064\,m^3$

a. $T_1 = ?$

H_2O @ 3 bar → Table A-3

$T_1 = \boxed{133.6°C}$

b. $T_2 = ?$

$v_2 = \frac{2.064\,m^3}{2\,kg}$

$= 1.032\,m^3/kg$

from table A-4

$\boxed{T_2 = 400°C}$

c. $W = P\Delta V$

$= (3E6\,Pa)(2.064\,m^3 - V_1)$

$V_1 = v_1 m$

$= (0.6058)(2)$

$= 1.2116\,m^3$

$= (3E6)(2.064 - 1.2116)$

$= \boxed{2,557\,kJ}$

255.7

4) Refrigerant 134a ; m = 1kg ; W = 20 kJ; KE & PE are negligible

State 1

$P_1 = 10\,bar$

$T_1 = 60°C$

$u_1 = 268.35\,kJ/kg$

State 2

$P_1 = 2\,bar$

$T_2 = 30°C$

$u_2 = 253.06\,kJ/kg$

a. $V_1 = ?$ from Table A-12

$v_1 = 0.02301\,m^3/kg$

$\boxed{V_1 = 0.02301\,m^3}$ $V = mv$

b. $Q - W = \Delta U + \cancel{KE}^0 + \cancel{PE}^0$

$Q = \Delta U + W$

$Q = (u_2 - u_1)m + W$

$= (253.06 - 268.35)(1) + (20)$

$= \boxed{4.71\,kJ}$

5) closed, rigid tank w/ 1.5 kg H_2O, State 1 = two-phase liquid-vapor mix @ $T_1 = 70°C$
Heat is transferred until State 2 = saturated vapor @ $T_2 = 120°C$

a. $P_1 = ?$ (kPa)
b. $Q = ?$ (kJ)

State 1	State 2
liquid vapor mix	sat. vapor
$T_1 = 70°C$	$T_2 = 120°C$
$V_1 = V_2$	$v_2 = v_g = 0.8857\ m^3/kg$
	$u_2 = u_g = 2529.5\ kJ/kg$

} Table A-3

from table A-3

	x	y
1	69.10	0.30
2	75.87	0.40

$$y = y_1 + (x - x_1)\frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= (0.30) + (70 - 69.10)\frac{(0.40 - 0.30)}{(75.87 - 69.10)}$$

$$= 0.3133 \text{ bar} \rightarrow \boxed{31.33\ kPa} = P_1$$

$$Q - W = \Delta U$$

$$Q = \Delta U + W$$

$$= (u_2 - u_1)m + (P\Delta V)$$

$\Delta V = 0$ b/c "rigid tank"

$$= m(u_2 - u_1)$$

$$= (1.5\ kg)(2529.5 - u_1)$$

u_1 = use $v_2 = v_1$ & v_f & v_g to get x
the use x to get u_1

$$v = v_f + x(v_g - v_f) \quad \text{(state 1)}$$

$v_f @ 70°C = 1.02285\text{E-3}\ m^3/kg$

$v_g @ 70°C = 5.0646\ m^3/kg$

$$(0.8857\ m^3/kg) = (1.02285\text{E-3}) + x(5.0646 - 1.02285\text{E-3})$$

$$x = 0.1746$$

$$u_1 = u_{1f} + x(u_{1g} - u_{1f})$$

$u_f @ 70°C = 292.966\ kJ/kg$

$u_g @ 70°C = 2469.54\ kJ/kg$

$$u_1 = (292.966) + (0.1746)(2469.54 - 292.966)$$

$$= 672.995\ kJ/kg$$

$$Q = m(u_2 - u_1)$$

$$Q = (1.5)(2529.5 - 672.9)$$

$$= \boxed{2784.75\ kJ}$$

6) liquid-vapor mix of Refrigerant 22 @ 10 bar ; $m_{sat\ liquid} = 25\ kg$; $x = 60\%$

a. Find V_{tank} (m^3)

b. Find fraction of V occupied by sat. vapor

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a. interpolating from Table A-7:

@ 10 bar $\begin{cases} v_f = 0.8352\text{E-}3\ m^3/kg \\ v_g = 0.0236\ m^3/kg \end{cases}$

$m_{sat.\ liquid} = m_f$

$v_f = \dfrac{V_f}{m_f}$

$V_f = m_f v_f$

$= (25\ kg)(0.8352\text{E-}3\ m^3/kg)$

$= 0.02088\ m^3$

$V_{tank} = V_f + V_g$

$= (m_f v_f) + (m_g v_g)$

$= (0.02088) +$

$m_g = ?$ $x = \dfrac{m_g}{m_f + m_g}$

$(0.6) = \dfrac{m_g}{(25) + m_g}$

$m_g = 37.5\ kg$

$= (0.02088) + ((37.5)(0.0236))$

$= \boxed{0.9058\ m^3}$

b. $V_{sat.\ vapor} = V_g = (37.5)(0.0236) = \boxed{0.885\ m^3}$

80%

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7) Balloon with 0.2 grams Helium ; assume ideal gas

$T_1 = 27°C$ $T_2 = 17°C$

$P_1 = 1.3\ bar$ $P_2 = 0.8$

from Table A-1, $M_{He} = 4.003\ kg/kmol$ (g/mol)

a. Determine R ($\frac{J}{g \cdot K}$)

$PV = mRT$

$R = \dfrac{\bar{R}}{M}$

$= \dfrac{8.314\ (J/mol\,K)}{4.003\ g/mol}$

$\boxed{R = 2.077\ J/g\,K}$

b. $V_i = ?$ (cm^3)

$P_1 V_1 = m R T_1$

$V_1 = \dfrac{m R T_1}{P_1}$

$= \dfrac{(2\text{E-}4)(2.077\ J/gK)(27°C + 273.15)}{(1.3\ \text{E}2\ kPa)}$

$= \boxed{958.6\ cm^3}$

c. $V_f = ?$ (cm^3)

$V_f = \dfrac{m R T_2}{P_2}$

$= \dfrac{(2\text{E-}4\ kg)(2.077\ J/gK)(27 + 273.15)}{(1.3\text{E}2\ kPa)}$

$= \boxed{1505.8\ cm^3}$

8) CO_2 in a closed, rigid tank with a paddle wheel

State 1	State 2
$T_1 = 25°C$	$T_2 = 257°C$
$P_1 = 2.75$ bar	$P_2 =$
$V_1 = 0.05 m^3$	$V_2 = 0.05 m^3$

Q from gas → surroundings = 2.75 kJ

Assume ideal gas

$PV = mRT$

a. Find m_{CO_2} in kg ⟶ $R = \frac{\bar{R}}{M}$; $\bar{R} = 8.314 \frac{kJ}{mol K}$; $M_{CO_2} = 44.01$ kg/kmol

(from Table A-1)

$R = \frac{(8.314)}{(44.01)} = 0.1889 \frac{kJ}{kg K}$

$\frac{PV}{RT} = m = \frac{(2.75E5 \, Pa)(0.05 m^3)}{(0.1889 \frac{kJ}{kg} K)(25°C + 273.15)}$

$= 244.12 \, g$ $\quad \frac{kJ}{kg} \to g$

$= \boxed{0.244 \text{ kg}}$

✓

b. Find W in kJ

↳ $Q + W = \Delta U + \cancel{\Delta KE}^{0} + \cancel{\Delta PE}^{0}$

$(-2.75 kJ) + W = m(u_2 - u_1)$

$u_2 @ T_2 = (257°C + 273.15) = 14{,}622 \, kJ/kg$

$u_1 @ T_1 = (25°C + 273.15) = 6{,}885 \, kJ/kg$

$W = (2.75 kJ) + (0.244 kg)(14{,}622 \frac{kJ}{kg} - 6{,}885 \frac{kJ}{kg})$

$= \boxed{1890.58 \text{ kJ}}$

8/0/6

$\bar{u} \neq u$

$\frac{\bar{u}}{M} = u$